

FORMULAS FOR REFERENCE

SPHERE	Surface area	$= 4\pi r^2$
	Volume	$= \frac{4}{3}\pi r^3$
CYLINDER	Area of curved surface	$= 2\pi rh$
	Volume	$= \pi r^2 h$
CONE	Area of curved surface	$= \pi rl$
	Volume	$= \frac{1}{3}\pi r^2 h$
PRISM	Volume	$= \text{base area} \times \text{height}$
PYRAMID	Volume	$= \frac{1}{3} \times \text{base area} \times \text{height}$

There are 36 questions in Section A and 18 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.

Section A

1. If $x = \frac{y(z-3)}{3z}$, then $z =$

A. $\frac{3}{3x-y}$

B. $\frac{-3}{3x-y}$

C. $\frac{3y}{3x-y}$

D. $\frac{-3y}{3x-y}$

E. $\frac{3x-y}{3y}$

2. If $f(x) = x^2 - 3x - 1$, then $f(a) + f(-a) =$

A. $2a^2$

B. $2a^2 - 2$

C. $6a$

D. $-6a$

E. -2

3. Solve $x^2 + 5x - 6 \leq 0$.

- A. $-6 \leq x \leq 1$
- B. $-3 \leq x \leq -2$
- C. $-1 \leq x \leq 6$
- D. $x \leq -6$ or $x \geq 1$
- E. $x \leq -1$ or $x \geq 6$

4. Solve the simultaneous equations:

$$\begin{cases} 2x + \frac{3}{y} = -1 \\ x - \frac{1}{y} = 7 \end{cases}$$

- A. $(0, -3)$
- B. $(1, -1)$
- C. $(4, -\frac{1}{3})$
- D. $(4, -3)$
- E. $(22, -\frac{1}{15})$

5. If $(x+3)^2 - (x+1)(x-3) \equiv P(x+1) + Q$, find P and Q .

- A. $P=2, Q=4$
- B. $P=2, Q=10$
- C. $P=4, Q=2$
- D. $P=4, Q=8$
- E. $P=8, Q=4$

6. Let $f(x) = 2x^3 - x^2 - 7x + 6$. It is known that $f(-2) = 0$ and $f(1) = 0$. $f(x)$ can be factorized as

- A. $(x+1)(x+2)(2x-3)$.
- B. $(x+1)(x-2)(2x+3)$.
- C. $(x-1)(x+2)(2x+3)$.
- D. $(x-1)(x+2)(2x-3)$.
- E. $(x-1)(x-2)(2x+3)$.

7. $\frac{(2^m)^2}{8^m} =$

- A. $\frac{2}{3}$
- B. 2^{-m}
- C. 2^m
- D. 2^{m^2-3m}
- E. 2^{2m^2-3m}

8. Factorize $x^2 - y^2 + 2x + 1$.

- A. $(x + y + 1)(x + y - 1)$
- B. $(x + y + 1)(x - y + 1)$
- C. $(x + y - 1)(x - y + 1)$
- D. $(x + y - 1)(x - y - 1)$
- E. $(x - y + 1)(x - y - 1)$

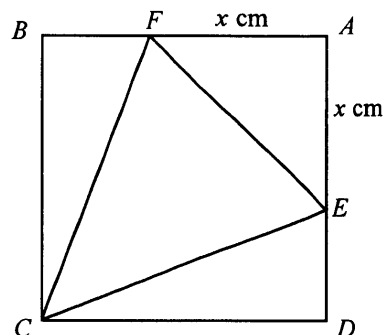
9. If the equation $x^2 - 6x + k = 0$ has real roots, find all possible values of k .

- A. $k \geq 9$
- B. $k \geq -9$
- C. $k = 9$
- D. $k \leq 9$
- E. $k \leq -9$

10. Solve $(x - 1)(x - 3) = x - 3$.

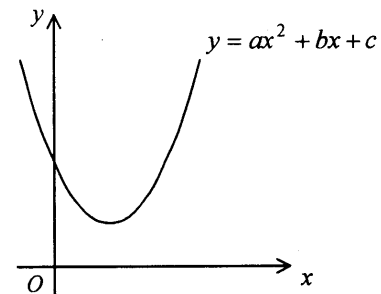
- A. $x = 1$
- B. $x = 2$
- C. $x = 0$ or 3
- D. $x = 1$ or 3
- E. $x = 2$ or 3

11. In the figure, $ABCD$ is a square of side 10 cm. If $AE = AF$ and the area of $\triangle CEF$ is 20 cm^2 , which of the following equations can be used to find AF ?



- A. $x^2 + 10(10 - x) + 20 = 100$
 B. $x^2 + 20(10 - x) + 20 = 100$
 C. $\frac{1}{2}x^2 + 10x + 20 = 100$
 D. $\frac{1}{2}x^2 + 10(10 - x) + 20 = 100$
 E. $\frac{1}{2}x^2 + \frac{10(10 - x)}{2} + 20 = 100$

12. The figure shows the graph of $y = ax^2 + bx + c$. Which of the following is true?



- A. $a > 0$, $c > 0$ and $b^2 - 4ac > 0$
 B. $a > 0$, $c > 0$ and $b^2 - 4ac < 0$
 C. $a > 0$, $c < 0$ and $b^2 - 4ac < 0$
 D. $a < 0$, $c > 0$ and $b^2 - 4ac > 0$
 E. $a < 0$, $c < 0$ and $b^2 - 4ac > 0$
13. If a , b , c , d are consecutive terms of an arithmetic sequence, which of the following must be true?
- I. $b - a = d - c$
 II. d, c, b, a are consecutive terms of an arithmetic sequence
 III. $a < b < c < d$
- A. I only
 B. I and II only
 C. I and III only
 D. II and III only
 E. I, II and III

14. A man bought a box of 200 apples for \$ 500 . 10 of the apples were rotten and the rest were sold at \$ 4 each. Find his percentage profit correct to 2 significant figures.

A. 34%
B. 38%
C. 52%
D. 57%
E. 60%

15. If $\frac{x+2y}{3x-4y} = 5$, then $x : y =$

A. 3 : 7 .
B. 7 : 3 .
C. 7 : 11 .
D. 9 : 7 .
E. 11 : 7 .

16. If $\frac{a}{b} = \frac{c}{d}$, which of the following must be true?

I. $\frac{a}{c} = \frac{b}{d}$
II. $\frac{a+b}{b} = \frac{c+d}{d}$
III. $\frac{a-b}{b} = \frac{c-d}{d}$

A. I only
B. I and II only
C. I and III only
D. II and III only
E. I, II and III

17. If x varies inversely as y and directly as z^2 , then

A. $\frac{x}{yz^2}$ is a constant.

B. $\frac{xy}{z^2}$ is a constant.

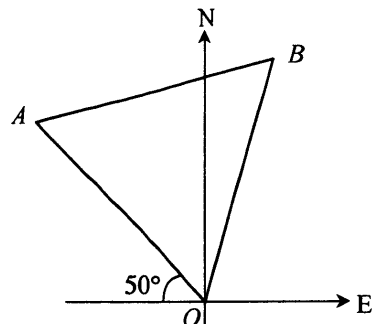
C. $\frac{xz^2}{y}$ is a constant.

D. $\frac{z^2}{y}$ is a constant.

E. $\frac{1}{y} + z^2$ is a constant.

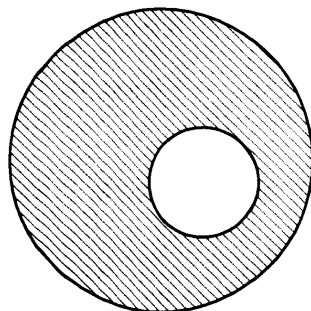
18. In the figure, OAB is an equilateral triangle. Find the bearing of B from A .

- A. 10°
- B. 80°
- C. 170°
- D. 260°
- E. 350°



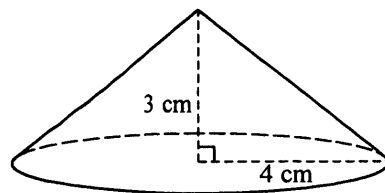
19. In the figure, the radii of the two circles are 3 cm and 1 cm respectively. Find the ratio of the area of the shaded part to that of the smaller circle.

- A. 2 : 1
- B. 3 : 1
- C. 4 : 1
- D. 8 : 1
- E. 9 : 1



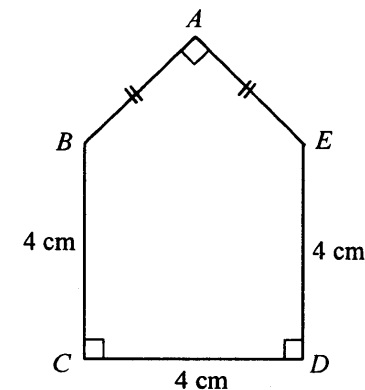
20. The figure shows a right circular cone of base radius 4 cm and height 3 cm. Find the area of its curved surface.

- A. $12\pi \text{ cm}^2$
- B. $16\pi \text{ cm}^2$
- C. $20\pi \text{ cm}^2$
- D. $24\pi \text{ cm}^2$
- E. $48\pi \text{ cm}^2$



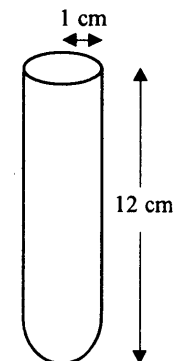
21. In the figure, find the area of the pentagon $ABCDE$.

- A. 16 cm^2
- B. 18 cm^2
- C. 20 cm^2
- D. 24 cm^2
- E. 32 cm^2



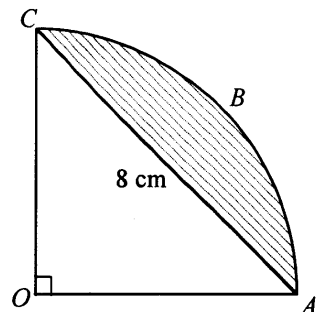
22. The figure shows a test tube consisting of a cylindrical upper part of radius 1 cm and a hemispherical lower part of the same radius. If the height of the test tube is 12 cm, find its capacity.

- A. $\frac{35}{3}\pi \text{ cm}^3$
- B. $\frac{37}{3}\pi \text{ cm}^3$
- C. $\frac{38}{3}\pi \text{ cm}^3$
- D. $\frac{40}{3}\pi \text{ cm}^3$
- E. $\frac{68}{3}\pi \text{ cm}^3$



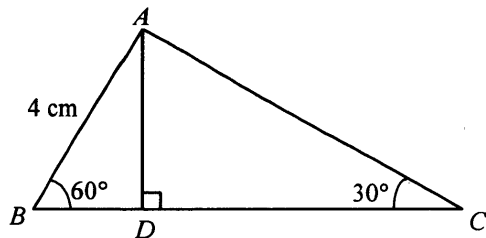
23. In the figure, $OABC$ is a sector. Find the area of the shaded region.

- A. $(\pi - 2) \text{ cm}^2$
 B. $(2\pi - 4) \text{ cm}^2$
 C. $(4\pi - 8) \text{ cm}^2$
 D. $(8\pi - 8) \text{ cm}^2$
 E. $(8\pi - 16) \text{ cm}^2$



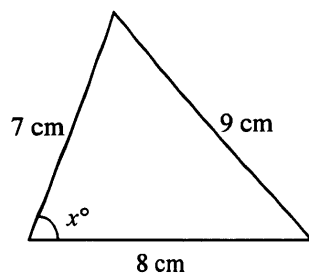
24. In the figure, find CD .

- A. 6 cm
 B. 4 cm
 C. $4\sqrt{3}$ cm
 D. $2\sqrt{3}$ cm
 E. $\frac{2\sqrt{3}}{3}$ cm



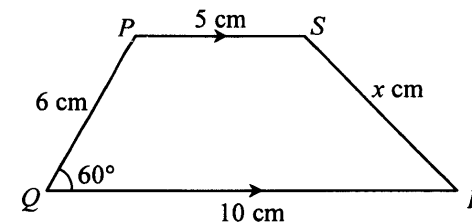
25. In the figure, find x correct to 3 significant figures.

- A. 48.2
 B. 55.1
 C. 58.4
 D. 67.5
 E. 73.4



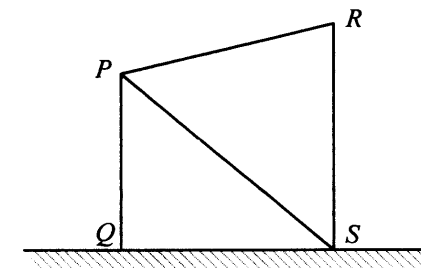
26. In the figure, $PQRS$ is a trapezium. Find x correct to 3 significant figures.

- A. 3.01
 B. 5.57
 C. 5.77
 D. 6.00
 E. 9.54



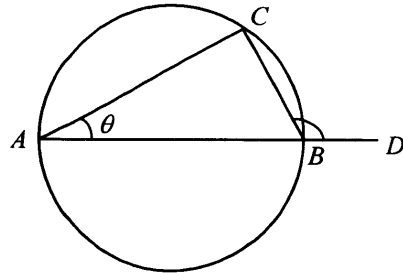
27. In the figure, PQ and RS are two vertical poles standing on the horizontal ground. The angle of elevation of R from P is 20° and the angle of depression of S from P is 40° . If $RS = 5 \text{ m}$, then $PR =$

- A. $\frac{5 \sin 40^\circ}{\sin 70^\circ} \text{ m}$.
 B. $\frac{5 \sin 50^\circ}{\sin 60^\circ} \text{ m}$.
 C. $\frac{5 \sin 60^\circ}{\sin 50^\circ} \text{ m}$.
 D. $\frac{5 \sin 70^\circ}{\sin 40^\circ} \text{ m}$.
 E. $\frac{5}{\sin 50^\circ \sin 60^\circ} \text{ m}$.



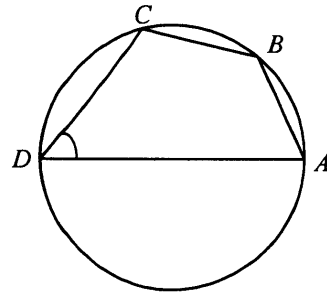
28. In the figure, AB is a diameter of the circle and ABD is a straight line. $\angle CBD =$

- A. 2θ .
- B. 4θ .
- C. $90^\circ + \theta$.
- D. $180^\circ - \theta$.
- E. $180^\circ - 2\theta$.



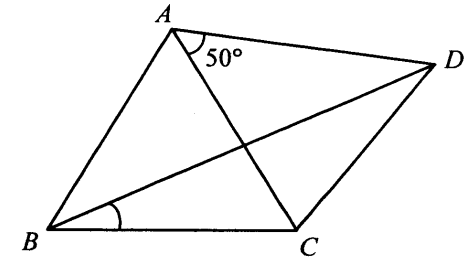
29. In the figure, AD is a diameter of the circle. If $\widehat{AB} : \widehat{BC} : \widehat{CD} = 3 : 5 : 7$, then $\angle ADC =$

- A. 36° .
- B. 45° .
- C. 48° .
- D. 49° .
- E. 72° .



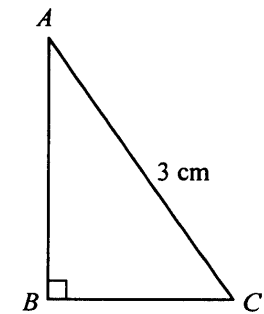
30. In the figure, $AB = BC = CA = CD$. Find $\angle CBD$.

- A. 20°
- B. 25°
- C. 27.5°
- D. 30°
- E. 35°



31. In the figure, $AB = 2BC$. Find BC correct to 3 significant figures.

- A. 0.775 cm
- B. 1.00 cm
- C. 1.34 cm
- D. 1.73 cm
- E. 1.80 cm

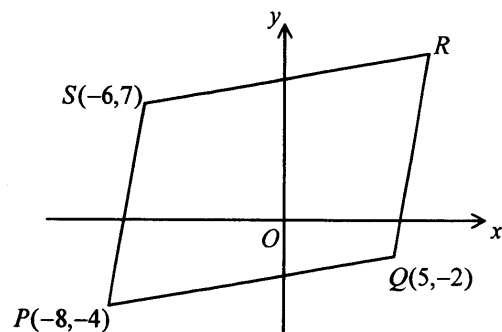


32. Find the equation of the straight line passing through $(-1, 1)$ and parallel to $5x + 4y = 0$.

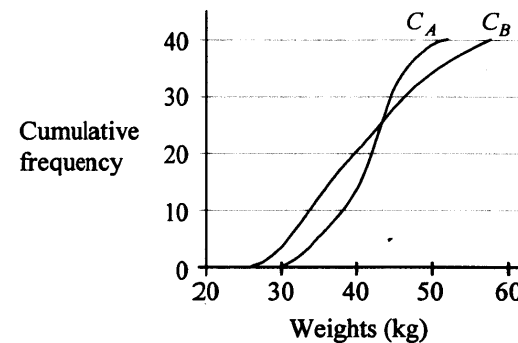
- A. $4x - 5y + 9 = 0$
- B. $4x + 5y + 1 = 0$
- C. $5x - 4y + 9 = 0$
- D. $5x + 4y - 1 = 0$
- E. $5x + 4y + 1 = 0$

33. In the figure, $PQRS$ is a parallelogram. Find the slope of PR .

- A. $\frac{13}{15}$
- B. $\frac{15}{13}$
- C. $\frac{9}{11}$
- D. $\frac{11}{9}$
- E. -5



34. In the figure, C_A and C_B are the cumulative frequency curves of two distributions of weights A and B respectively. Which of the following is/are true?



- I. median of $A >$ median of B
 - II. range of $A >$ range of B
 - III. inter-quartile range of $A >$ inter-quartile range of B
- A. I only
 - B. I and II only
 - C. I and III only
 - D. II and III only
 - E. I, II and III

35. Two cards are drawn randomly from five cards numbered 2, 2, 3, 5 and 5, respectively. Find the probability that the sum of the numbers on the cards drawn is 5.

- A. $\frac{1}{5}$
- B. $\frac{2}{5}$
- C. $\frac{1}{10}$
- D. $\frac{2}{25}$
- E. $\frac{4}{25}$

36. In a shooting game, the probability that Mr. Tung will hit the target is $\frac{2}{3}$. If he shoots twice, find the probability that he will hit the target at least once.

- A. $\frac{1}{9}$
- B. $\frac{2}{9}$
- C. $\frac{4}{9}$
- D. $\frac{2}{3}$
- E. $\frac{8}{9}$

Section B

37. Let a and b be two consecutive positive integers. Which of the following must be true?

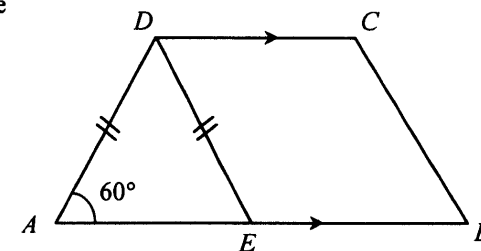
- I. $a + b$ is odd.
- II. ab is odd.
- III. $a^2 + b^2$ is odd.

- A. III only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

38. In the figure, $ABCD$ is a trapezium. Which of the following must be true?

- I. AED is an equilateral triangle
- II. $EBCD$ is a parallelogram
- III. $AB = 2DC$

- A. I only
- B. II only
- C. I and II only
- D. I and III only
- E. I, II and III



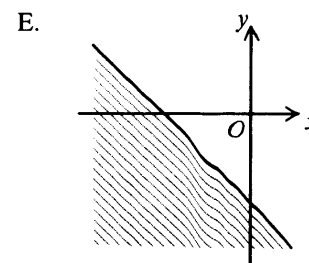
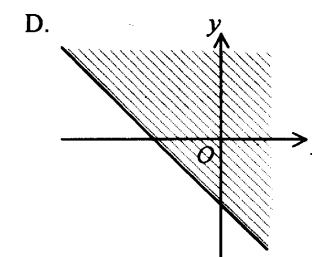
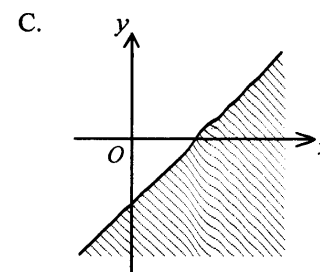
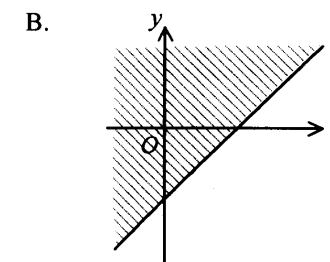
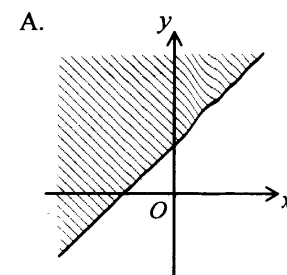
39. $\frac{2}{x^2-1} - \frac{3}{x^2-x-2} =$

- A. $\frac{-1}{(x-1)(x-2)}$
 B. $\frac{-1}{(x+1)(x-2)}$
 C. $\frac{-1}{(x+1)(x+2)}$
 D. $\frac{-1}{(x-1)(x+1)(x-2)}$
 E. $\frac{-x-7}{(x-1)(x+1)(x-2)}$

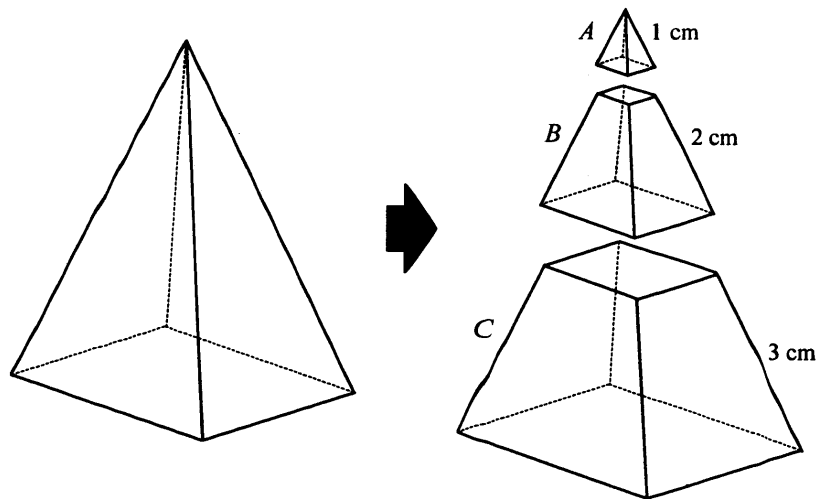
40. Suppose $\log_{10} 2 = a$ and $\log_{10} 3 = b$. Express $\log_{10} 15$ in terms of a and b .

- A. $-a+b+1$
 B. $-a+10b$
 C. $a+2b$
 D. $(a+b)b$
 E. $\frac{10b}{a}$

41. If $b < 0$ and $c < 0$, which of the following shaded regions may represent the solution of $x + by + c \geq 0$?



42. In the figure, a right pyramid with a square base is divided into three parts A , B and C by two planes parallel to the base such that the lengths of their slant edges are 1 cm, 2 cm and 3 cm respectively.



Find volume of A : volume of B : volume of C .

- A. 1 : 2 : 3
- B. 1 : 4 : 9
- C. 1 : 8 : 27
- D. 1 : 26 : 189
- E. 1 : 27 : 216

43. Find the sum to infinity of the geometric sequence $-1, \frac{1}{x}, -\frac{1}{x^2}, \frac{1}{x^3}, \dots$, where $x > 1$.

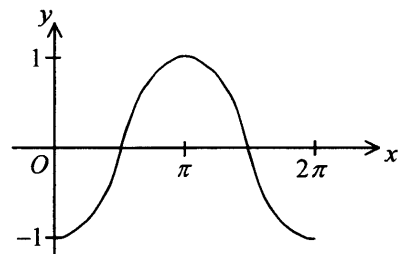
- A. $\frac{-1}{x-1}$
- B. $\frac{-1}{x+1}$
- C. $\frac{-x}{x-1}$
- D. $\frac{-x}{x+1}$
- E. $\frac{x}{x+1}$

44. $\frac{1+\sin \theta}{\cos \theta} + \frac{\cos \theta}{1+\sin \theta} =$

- A. 1.
- B. $2(1+\sin \theta)$.
- C. $\frac{2}{\cos \theta}$.
- D. $\frac{2}{\cos \theta(1+\sin \theta)}$.
- E. $\frac{1+\sin \theta+\cos \theta}{\cos \theta(1+\sin \theta)}$.

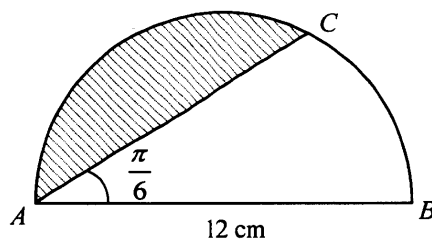
45. The figure shows the graph of the function

- A. $y = \cos x$.
 B. $y = \cos(-x)$.
 C. $y = \cos(\frac{\pi}{2} - x)$.
 D. $y = \cos(\frac{\pi}{2} + x)$.
 E. $y = \cos(\pi - x)$.



46. In the figure, ABC is a semicircle. Find the area of the shaded part.

- A. $6\pi \text{ cm}^2$
 B. $15\pi \text{ cm}^2$
 C. $(6\pi - 9\sqrt{3}) \text{ cm}^2$
 D. $(6\pi + 9\sqrt{3}) \text{ cm}^2$
 E. $(12\pi - 9\sqrt{3}) \text{ cm}^2$

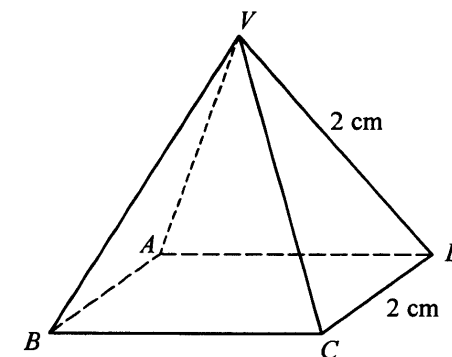


47. For $0^\circ \leq x \leq 360^\circ$, how many roots does the equation $3 \sin^2 x + 2 \sin x - 1 = 0$ have?

- A. 0
 B. 1
 C. 2
 D. 3
 E. 4

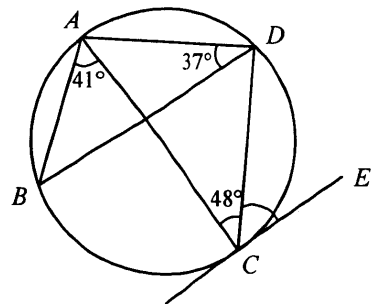
48. The figure shows a right pyramid with a square base $ABCD$. Let θ be the angle between the planes VAB and VCD . Find $\sin \frac{\theta}{2}$.

- A. $\frac{1}{2}$
 B. $\frac{\sqrt{3}}{2}$
 C. $\frac{1}{\sqrt{3}}$
 D. $\frac{1}{\sqrt{5}}$
 E. $\frac{2}{\sqrt{5}}$



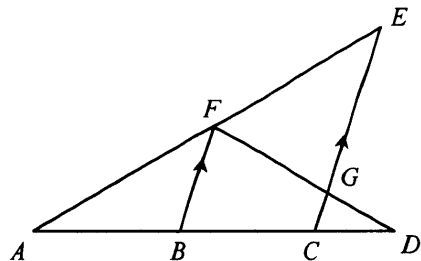
49. In the figure, CE is tangent to the circle at C . Find $\angle DCE$.

- A. 40°
- B. 42°
- C. 49°
- D. 54°
- E. 78°



50. In the figure, $ABCD$, AFE , CGE and FGD are straight lines. If $AB = BC = 2CD$, then $CG : GE =$

- A. $1 : 2$.
- B. $1 : 3$.
- C. $1 : 4$.
- D. $1 : 5$.
- E. $1 : 6$.



51. Find the mean deviation of the five numbers $x-2$, $x-1$, x , $x+1$ and $x+2$.

- A. x
- B. 0
- C. $\frac{6}{5}$
- D. $\sqrt{2}$
- E. $\frac{\sqrt{30}}{5}$

52. The circle $x^2 + y^2 - 2x - 7y - 8 = 0$ intersects the x -axis at A and B . Find the length of AB .

- A. 2
- B. 6
- C. 7
- D. 9
- E. $\sqrt{85}$

53. The equations of two circles are

$$x^2 + y^2 + ax - by = 0 \text{ and}$$

$$x^2 + y^2 - ax + by = 0 .$$

Which of the following must be true?

- I. The two circles have the same centre.
- II. The two circles have equal radii.
- III. The line joining the centres of the two circles passes through the origin.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

54. $A(7, 14)$ and $B(1, 2)$ are two points. C is a point on AB produced such that $AB : BC = 2 : 1$. Find the coordinates of C .

- A. $(-5, -10)$
- B. $(-2, -4)$
- C. $(3, 6)$
- D. $(5, 10)$
- E. $(10, 20)$

END OF PAPER